



Device:

Topic	Function
Power	Power Estimates
Power	Power Delivery Solution
Power	Voltage Assignments (MCU)
Power	
Power	
Power	
Power	
Power	
Power	
Power	
Power	
Power	
Power	
Power	Voltage Assignments (Main)
Power	
Power	

[illegible]

Power	
Power	
Power	
Power	Voltage Assignments (LPDDR4)
Power	
Power	
Power	Analog Filters
Power	
Power	
Power	Internal LDO Caps
Power	Decoupling Caps
Power	
Power	IO Levels
Power	
Power	

Power	
Power	Power Sequence
Power	
Power	
Power	Low Power Modes
Power	
Power	
Power	Power Management IC (PMIC)
Power	
Power	

Power	
Power	
Power	
Power	
Power	
Reset	Power On Reset
Reset	
Reset	
Reset	
Reset	
Reset	Warm Reset
Reset	
Reset	Reset Output
Reset	
Reset	
Clock	Main System Clock
Clock	
Clock	Optional HF Clock
Clock	

Clock	Option LF Clock
Clock	
Clock	Crystal Clock Source
Clock	
Clock	Oscillator Clock Source
BootMode	BootMode Control
BootMode	Boot Mode Selection
BootMode	
BootMode	
BootMode	
BootMode	
BootMode	
BootMode	
BootMode	
BootMode	
BootMode	
BootMode	High Security (HS) Device
Unused	Mandatory External Pull-Down
Unused	
Unused	
Unused	
Unused	Mandatory External Pull-Up
Unused	
Unused	
Unused	
Unused	Power Supply
Unused	Do not exist
Unused	Reserved Pins

Unused	
LPDDR4	Control Signals (Multi Point)
LPDDR4	
LPDDR4	
LPDDR4	Control Signals (Point-Point)
LPDDR4	
LPDDR4	Data Signals
LPDDR4	
LPDDR4	

LPDDR4	
LPDDR4	IO Calibration
LPDDR4	Dual Rank
LPDDR4	Retention (Suspend-to-RAM)
LPDDR4	Memory Reset
LPDDR4	Other Checks
LPDDR4	
LPDDR4	
LPDDR4	
LPDDR4	
LPDDR4	
OSPI	Interface Connections
OSPI	

OSPI	
OSPI	
OSPI	Bootable Interface
OSPI	
OSPI	IO Voltage Level
OSPI	Other
High Speed Serial	IO Calibration
High Speed Serial	
High Speed Serial	
High Speed Serial	Multiple Protocols
High Speed Serial	
High Speed Serial	USB Connections
High Speed Serial	
High Speed Serial	

High Speed Serial	
High Speed Serial	
High Speed Serial	
High Speed Serial	
High Speed Serial	PCIe Connections
High Speed Serial	
High Speed Serial	
High Speed Serial	
High Speed Serial	SGMII Connections
High Speed Serial	
High Speed Serial	
High Speed Serial	
General	Debug
General	
General	

General	
General	
General	
General	High Security (HS) Device
General	I2C
General	
General	MMC
General	
General	
General	
General	
General	Safety
General	Monitor Signals
General	SysConfig Tool

AM6x/TDA4x/DRA8x Schematic

AM69

Item
Have you used the output of the processor's power model along with estimates from the rest of your design to determine the power solution needed?
Check that the design utilizes one of the recommended and supported power delivery solution/schemes defined for this processor.
Have you checked that the correct voltage(s) are applied to the MCU VDD power rail to the device?
Review with PDN (identified in other checklist entry) to ensure rail sourced from correct supply
Have you checked that the correct voltage(s) are applied to the MCU Analog power rail to the device?
Review with PDN (identified in other checklist entry) to ensure rail sourced from correct supply
Have you checked that the correct voltage(s) are applied to the MCU IO power rails to the device (MCU_VDDSHVn)?
Review with PDN (identified in other checklist entry) to ensure rail sourced from correct supply. As peripheral loading will vary from design/design, take caution to not overload source.
Have you checked that the correct voltage(s) are applied to the MCU IO Retention power rail to the device?
Review with PDN (identified in other checklist entry) to ensure rail sourced from correct supply. Different connection may exist depending on if low power feature is used.
Have you checked that the correct voltage(s) are applied to the CPU VDD power rail to the device?
Review with PDN (identified in other checklist entry) to ensure rail sourced from correct supply
Have you checked that the correct voltage(s) are applied to the Core VDD power rail to the device?
Review with PDN (identified in other checklist entry) to ensure rail sourced from correct supply
Have you checked that the correct voltage(s) are applied to the low voltage PHY power rail to the device?

Critical: These power rails must be sourced from same supply as VDD_CORE. There are filter components isolating from VDD_CORE, but must be sourced from same supply.

Review with PDN (identified in other checklist entry) to ensure rail sourced from correct supply

Have you checked that the correct voltage(s) are applied to the SRAM power rail to the device?

Review with PDN (identified in other checklist entry) to ensure rail sourced

Have you checked that the correct voltage(s) are applied to the CORE IO Retention power rail to the device?

Review with PDN (identified in other checklist entry) to ensure rail sourced from correct supply. Different connection may exist depending on if low power feature is used.

Have you checked that the correct voltage(s) are applied to the analog low voltage PLL/DLL power rail to the device?

Review with PDN (identified in other checklist entry) to ensure rail sourced

Have you checked that the correct voltage(s) are applied to the 1.8V PLL/DLL power rail to the device?

Review with PDN (identified in other checklist entry) to ensure rail sourced from correct supply

Have you checked that the correct voltage(s) are applied to the USB2.0 PHY power rail to the device?

Review with PDN (identified in other checklist entry) to ensure rail sourced from correct supply

Have you checked that the correct voltage(s) are applied to the IO power rails to the device (non-MCU)?

Review with PDN (identified in other checklist entry) to ensure rail sourced from correct supply. As peripheral loading will vary from design/design,

Have you checked that the correct voltage(s) are applied to the MMC0 IO power rail to the device?

Review with PDN (identified in other checklist entry) to ensure rail sourced from correct supply. As peripheral loading will vary from design/design, take caution to not overload source.

Have you checked that the correct voltage(s) are applied to the MMC1 IO power rail to the device?
Review with PDN (identified in other checklist entry) to ensure rail sourced from correct supply. Optionally, if compliant dual voltage SD Card operation is desired, then this IO domain can be source from dual-voltage (change dynamically) regulator.
Have you checked that the correct voltage(s) are applied to the eFuse power rail to the device? (Optional)
Review with PDN (identified in other checklist entry) to ensure rail sourced from correct supply. Only Required for High Secure (HS) devices to fuse custom key(s).
If supporting VPP supply for setting of eFuse (optional), have you verified the increased current requirement can be supported?
Have you checked that the correct voltage(s) are applied to the DDR IO power rail to the device?
Review with PDN (identified in other checklist entry) to ensure rail sourced from correct supply. As LPDDR rails (VDDQ, VDD2) loading will vary from
Nominal voltage is 1.80, Voltage rails for LPDDR4 Memory only (no SoC connection)
SoC contains multiple analog power pins (VDDA_x) that provide power to sensitive analog circuitry such as PLLs, DLLs, and SERDES drivers/recievers. Do each of these use seperate filters when connecting to their power sources?
Does the 0.8V PLL/DLL power rail (VDA_DLL_0V8) include the latest filter and minimum capacitance values?
Does the SRAM filter(s) support the current capacities from latest power estimates? Upated power estimates for VDDAR_MCU, VDDAR_CPU, and VDDAR_CORE may require a change in the package size for the filter component(s).
SoC includes internal LDOs which require connection to external capacitor(s). Do you have the proper capacitor connection and value for each of the internal LDOs?
Verify the power supply decoupling is adequate. All processor power rails should utilize both bulk and high frequency decoupling capacitors. The critical power domains that require the most attention are the low volatge, high current domains (MPU, CORE, CPU, etc).
(Schematic + PCB Deisgn) Has both static and dynamic power delivery analysis been performed on critical power rails? Verify the PDN targets for effective resistance (Reff), Capacitor loop inductance (Cap LL), and Target Impedance (TargZ) meet or exceed the recommended values.
Verify all IO pins on each VDDSHVn (or MCU_VDDSHVn) supply only supports single voltage.
Verify all peripheral IO pins match the voltage level of the processor's IO pins.
Verify all Pull-up resistors are connected to the same supply rail as the Processor's IO.

Have you verified voltage will not be supplied to processor IOs prior to powering VDDSHVx power rail?
Does your design meet the required power sequencing documented in the data manual? Proper power supply sequencing also includes proper correlation with resets and primary clocks.
Does your design provide power to all power pins? If portions of the processor are unused in your design, each power rail must still required to be supplied with proper power levels.
If you design uses custom eFuse programming, verify the VPP power sequence is correct. Only Required for High Secure (HS) devices to fuse custom key(s).
Does the design support MCU-Only low power mode? For this the MCU can continue operations when the Main portion of SoC are powered-down to reduce system power. Entering/Existing MCU-Only requires SoC Main voltage domains to power up/down according to recommended power down seq.
Does design support IO Retention Mode? For this the RETENTION domains (a specialized CORE and specific IO rail) can continue to operate when the rest of the SoC is powered-down to minimize system power. Exit from IO Retention mode is implemented with detection of change on IO, and WAKE event being generated from SoC to the Power Management IC (PMIC).
Does design support Suspend To RAM (S2R) low power mode? For this the SoC's EMIF/DDR and external SDRAM component voltage domains remain energized when all other SoC voltage domains are powered-down to reduce system power. Entry into S2R mode is typically implemented via software issuing I2C commands controlling PMIC power & GPIO resources accordingly. Processor MCU and Main voltage domains must power-up/down according to recommended power up seq.
Are the correct Power Management IC (PMIC) devices used in the design, and are they latest revision?
Does design expect to use TI's software for PMIC control? For most cases, designs should align with TI's EVMs for PMIC control resources.
Does design support PMIC interrupt connecting with processor IO?

Verify the PMIC's power rail assignments are correct according to latest power solution recommendation. Each PMIC is specifically programmed for a power solution (SMPSSs, LDOs, etc). PMICs can have different supported configurations, so ensure each PMIC regulator in design are validated against correct power solution.
Verify the PMIC's IO are connected correctly according to latest power solution recommendation. Each PMIC is specifically programmed for a power solution (resets, external regulator enables, etc). PMICs can have different supported configurations, so ensure design IO are validated against correct power solution.
(Schematic + PCB Design) Does the design support remote sensing for the PMIC SMPS regulators? Are the SMPSx feedback pins connected through separated connection to the point of load?
Some 3 or 4-phase PMIC SMPS regulators can support alternative voltage monitoring of an external power rail on the unused remote sense inputs (for Functional Safety needs).
Does the PMIC have a dedicated design checklist? Often power components will have design check items separate from this checklist.
Verify that the MCU Power-On Reset signal is properly implemented in the system. Signal asserted (low) while the SoC supplies are ramping up or down is mandatory.
Verify that the level of the signal applied to MCU_PORZ corresponds to the VDDA_WKUP voltage.
Verify that the Power-On Reset signal is properly implemented in the system. Signal asserted (low) while the SoC supplies are ramping up or down is mandatory.
Verify that the level of the signal applied to Power on Reset corresponds to the VDDA_WKUP voltage.
If you are using controller other than TI PMIC, do you have a reset circuit that generates the proper reset generation?
Verify the Warm Reset Signal(s) are properly implemented in the system. Warm reset is an optional function, and should be pulled-up if unused.
Verify warm reset voltage level matches the correct IO power domain.
Verify the Reset Output signal(s) are properly connected. It is recommended to use the RESETSTATz output for resetting all system peripherals that need to reset when the processor undergoes any type of global reset.
Verify reset output signal voltage level matches the correct IO power domain.
Verify peripheral resets controlled from processor are kept while the processor supplies are still ramping.
Does design have a clock source for WKUP_OSC0 (either crystal for oscillator)? This clock is required, and must be present for the device to operate.
Verify that the clock frequency on WKUP_OSC0 is supported by the device, and matches the setting of MCU_BOOTMODE pins.
Does design have a clock source for OSC1 (either crystal for oscillator)? This clock is NOT required, but can optionally be present for the if design requires other frequencies not supported from WKUP_OSC0 frequency.
If OSC1 is not used, do you have it properly terminated?

Does design have a clock source for low frequency clock (32.768-kHz)? This clock is NOT required, but can optionally be present for the if design requires low frequency/sleep clock.
Verify LF_CLKIN signal input voltage level matches the correct IO power domain.
If using crystal for source clock(s), does crystal design have proper Load Capacitors, ESR Ratings, Start-up Time, etc?
Does crystal design include Rd and Rbias resistors? It is optional, but preferable to have bias and dampening resistors that can help tune the crystal later in early version of the design.
If using oscillator for source clock(s), does it meet the device requirements (voltage level, duty cycle, etc)?
Do all BOOTMODE and MCU_BOOTMODE pins have external pulling resistors or a circuit to drive the desired bootmode?
Verify the boot mode strapping is correctly set for Default clock/PLL configuration.
Verify the boot mode strapping is correctly set for Primary Boot (Pt A)
Verify the boot strapping is correctly set if MCU Only mode boot is required.
Verify the boot strapping is correctly set if Override mode boot is required.
Verify the boot strapping is correctly set for POST processing mode.
Verify the boot strapping is set correctly for Primary Boot Mode (Pt B)
Verify the boot strapping is set correctly for Backup Boot Mode
Verify the boot strapping is set correctly for to set configuration options for Primary Boot Mode
Verify the boot strapping is set correctly for to set configuration options for Backup Boot Mode
Is the expected bootmode present on the BOOTMODE and MCU_BOOTMODE pins when PORz or MCU_PORz are active?
If design is supporting high security (HS) devices, it is mandatory to implement a method for booting / loading the keywriter image before a signed image can be booted.
If unused, Each of these balls must be connected to VSS through a separate external pull resistor to ensure these balls are held to a valid logic low level
If unused, Each of these balls must be connected to the corresponding power supply through a separate external pull resistor to ensure these balls are held to a valid logic high level
If unused, Each of these balls must be left unconnected.
These balls do not exist on the package
These balls must be left unconnected.

These balls must be left unconnected.

Does the design have correct LPDDR4 connections for differential clock?

Does design have correct LPDDR4 connections for Control/Address (CA) signals?

Does design have correct LPDDR4 connections for Clock Enable signals?

Does the design have correct LPDDR4 connections for chip select 0?

Does the design have correct LPDDR4 connections for chip select 1? May be left open if 2nd chip select not used (single rank)

Does the design have correct LPDDR4 connections for Data Byte 0?

Does the design have correct LPDDR4 connections for Data Byte 1?

Does the design have correct LPDDR4 connections for Data Byte 2?

Does the design have correct LPDDR4 connections for Data Byte 3?
Does the design have the LPDDR4 IO calibration set correctly?
Does the LPDDR4 memory device support dual rank? Dual rank is supported, but only if multiple die are within a single package. Multiple memory packages are NOT supported.
Does design support LPDDR4 Retention mode (Suspend-to-RAM)? This low power mode allows memory contents to stay in self refresh mode while most of processor is powered down.
Does the design have correct LPDDR4 connection for reset? Pulling the reset high (prior to processor driving reset low) may not result in a functional issue as such but may affect the life of the LPDDR4 memory device (due to in-rush current during Power-on.)
Verify the LPDDR4 Memory device IO termination is properly set.
Verify the LPDDR4 Memory device IO calibration is properly set.
Does design support LPDDR4X, which supports and IO voltage of 0.6V (rather than the standard 1.1V IO voltage)?
Do you have notes in you schematic or guidelines to ensure that the DDR routing is done properly?
Does your design include sufficient amount of decoupling for the DDR interface?
Does your design use less than the maximum supported LPDDR interfaces available? (Ex: If device supports two interfaces (DDR0 and DDR1), but design is only using one interface. Other interface is unused.)
Does design have proper connections for serial flash interface clock?
Does design have proper connections for serial flash memory chip select(s)

Does design have proper connections for the serial flash memory data signals?

Does the selected serial flash memory device support data strobe (DQS)?

Is the serial flash memory planned to be used as a bootable memory?

Does the selected serial flash memory device' boot modes align with a mode supported by the processor?

Verify the I/O voltage level of the memory matches the voltage applied to the processor's I/O power pins. VDDSHV1_MCU.

Does the 8bit OSPI interface connect with a dual-Quad SPI devices?

Does the design include the correct IO calibration resistors for USB interfaces?

Does the design include the correct IO calibration resistors for SERDES?

Does the design include the correct IO calibration resistors for serial MIPI interfaces (DSI, CSI interface)?

Each SERDES instance has a limited set of reference clocks it can generate. This can limit the different protocols that can be used simultaneously on a SERDES instance.

A single Serdes interface can support a limited number of different protocols simultaneously.

Does design include support for USB2.0 signals?

Is design planning to use interface for USB3.x?

Has design connected to correct SERDES for USB3 operation?
Does design support Type C interface?
Does design include proper use of DC blocking capacitors?
Recommended avoiding micro-AB connector for designs that support Super-Speed.
Is design planning to use interface for PCIe?
Has design connected to correct SERDES signals for PCIe operation?
Does the design use internal or external reference clock?
Does design include proper use of DC blocking capacitors?
Is design planning to use interface for SGMII?
Is design planning to use interface for USXGMII?
Does design support both SGMII and RGMII interfaces?
Does design support a 10/100 SGMII PHY?
Have you added visibiity for internal clocks?
Does design include a Emulation (JTAG) connection?
Does design connect Emulation Return Clock (RTCK) signal correctly on emulator interface?

Does design use correct IO voltage levels for Emulation/JTAG signals?
Does design support Emulation/TRACE interface for collecting internal processor information, used for debugging purposes?
Does design include a UART interface reserved for terminal/console?
(UART) Does design support High Security (HS) Devices and utilize the Keywriter Utility?
(I2C) Does design have I2C Pull-up to correct voltage?
(I2C) Does design require fully compliant I2C IO?
Does design include a series resistor on the Clock to dampen reflections?
Does design include pull-up resistors for SD Card signals to proper IO voltage rail?
Does design support UHS-I type cards/devices?
Does design plan to boot from MMC device?
Does design properly power SD Card VDD?
Does design have any Functional Safety requirements? If so, verify the processors Safety Monitor/Error signals are properly connected to the PMIC/Safety Monitor device.
Verify processor's external voltage monitor pins are connected correctly.
Have you entered this design's information into TI's System Configuration Tool?

AM69 / J784S4 / TDA4AH / TDA4AP/ TDA4VH / TDA4VP Device Details

The power needed for each rail of the processor will vary based on the interfaces used and the environment in which it is operated. Processor's power requirements must be determined using the power model. See ti.com for additional information.

The processor includes many different power rails which must be powered with the correct voltage and correct sequence for proper operation. It is STRONGLY recommended all designs use one of the defined PDNs (Power Delivery Networks) for powering the processor, as implemented on the TI Reference Platforms (EVM or SK) for the specific device. These power solutions are specifically defined to work with the processor and provide several different options for power/cost targets.

When using one of the PDNs, be aware the SMPS/LDO default voltages, power up/down sequences, and GPIO pin assignment are defined as part of the Power Delivery solution. Check with the corresponding PDN User's Guide and/or the Power Management (PMIC) documentation for latest released details.

Nominal voltage for VDD_MCU is 0.85V.

Power Pins: L24, M23, M25, N26...

Nominal voltage is 0.85V.

Power Pins for VDDAR_MCU: M27, N24

Nominal voltage for VDDA_ADC0, VDDA_ADC1 is 1.80V. Power pins: M31, N30

Nominal voltage for VDDA_WKUP is 1.80V. Power pins: K31, L32

Nominal voltage for VDDA_POR_WKUP is 1.80V. Power pins: N28

Nominal voltage for VDDA_MCU_PLLGRP0 is 1.80V. Power pins: M28

Nominal voltage for VDDA_MCU_TEMP is 1.80V. Power pins: M26

Nominal voltage for VDDSHV0_MCU can be either 1.8 V or 3.3 V.

Power pins: H29, J28, K29 (Used for MCU GENERAL IO)

Nominal voltage for VDDSHV1_MCU can be either 1.8 V or 3.3 V.

Power pins: H25, J24, K25 (Used for MCU OSPI IO)

Nominal voltage for VDDSHV2_MCU can be either 1.8 V or 3.3 V.

Power pins: H27, J26, K27 (Used for MCU RGMII IO)

Nominal voltage for VDD_MCU_WAKE1 is 0.80V - 0.85V.

Power pins: L28

Nominal voltage for VDD_CPU is 0.80V - 0.85V.

Power pins: H21, H23, J20...

Nominal voltage for VDD_CORE is 0.80V.

Power pins: R24, R26, R28...

Nominal voltage for VDDA_OP8_SERDESx and VDDA_OP8_SERDESx_C is 0.80V

Power pins: AJ12, AK13, AJ15, AK14, AJ20, AJ21, AJ17, AJ18 and AH12, AH13, AH15, AH16, AH20, AG21, AG17, AH18

Nominal voltage for VDDA_OP8_UFS is 0.80V
Power pins: AH11

Nominal voltage for VDDA_OP8_DSITX and VDDA_OP8_CSIRX is 0.80V
Power pins: AJ24, AJ25 and AJ26, AK26, AJ28

Nominal voltage for VDDA_OP8_USB is 0.80V
Power pins: AK20

Are all VDDA_PHY_OP8 rails source from same supply as VDD_CORE?

Nominal voltage for VDDAR_CPU is 0.85V. Power pins: M21, N23, T15...

Nominal voltage for VDDAR_CORE is 0.85V.
Power pins: T25, AB27, AC24...

Nominal voltage for VDD_WAKE0 is 0.80V - 0.85V.
Power pins: U29

Nominal voltage for VDDA_OP8_DDR_PLLx is 0.80V.
Power pins: U11, M14, N11, M18

Nominal voltage for VDD_OP8_DLL_MMC0 is 0.80V.
Power pins: AE9

Nominal voltage for VDDS_OSC1 is 1.8V.
Power pins: N29

Nominal voltage for VDDA_PLLGRP_x is 1.8V.
Power pins: AA27, Y28, AG13, V14, R21, P12, P15, Y26, AG23, AA23, AB26

Nominal voltage for VDDA_TEMP_x is 1.8V.
Power pins: Y27, M12, W23, AE13, AD18

Nominal voltage for VDDA_1P8_UFS is 1.8V.
Power pins: AJ10

Nominal voltage for VDDA_1P8_USB is 1.8V.
Power pins: AK21

Nominal voltage for VDDA_1P8_DSITX and VDDA_1P8_CSIRX is 1.8V.
Power pins: AH24, AH25 and AH27, AH28, AH29, AJ29

Nominal voltage for VDDA_1P8_SERDES_x is 1.8V.
Power pins: AJ13, AJ14, AH21, AJ23, AH17

Nominal voltage for VDDA_3P3_USB is 3.3V.
Power pins: AJ19

Nominal voltage for VDDSHV0 can be either 1.8 V or 3.3 V.
Power pins: V30, V32, W31

Nominal voltage for VDDSHV2 can be either 1.8 V or 3.3 V.
Power pins: T30, T32, U31

Nominal voltage for VDDS_MMC0 is 1.8V.
Power pins: AF9, AG8, AG10, AH9

Nominal voltage for VDDSHV5 can be either 1.8 V or 3.3 V, and can change dynamically.
Power pins: P31, R30, R31 (Used for MMC1 pins)

Nominal voltage for VPP_MCU and VPP_CORE is 1.80v, should be left open/floating if feature not required.
Power pins: L29 and AA31

Recommendation is to support up to 400mA for VPP power rail (VPP_MCU + VPP_CORE). TI's EVMs may have a regulator supporting only 300mA.

Nominal voltage for VDDS_DDR and VDDS_DDR_C is 1.10V.
Power pins: A31, B1, H9... and T10, L15, M10, L17

Nominal voltage for LPDDR4 Memory (VDDQ, VDD2) is 1.1V.
Power pins: See LPDDR4 data sheet

Nominal voltage for LPDDR4 Memory (VDD1) is 1.8V
Power pins: See LPDDR4 data sheet

Strongly recommend using same analog filters used in TI's latest EVM design, as these have been validated.
For example, the 0.8V SERDES power pins require ferrite bead + 22uF + 1uF + (0.1uF per power pin).

Provision analog supply filters as follows: 1x 0402 in-line component (0-ohm Resistor or Ferrite Bead) + 0805 cap to Gnd (22/100nF) + 0402 cap to Gnd (1.0uF) + 0201 high-freq decoupling cap to Gnd (0.1uF) at all power balls. This configuration allows flexibility to optimize filtering with different configurations (i.e. DNI, different cap values, 0-ohm resistors or ferrite beads).

See TI's latest EVM design for filter component recommendations. The recommendation is to limit the IR drop of filter to 1%.

Each CAP_VDDsx and CAP_VDDsx_MCU pin requires separate 1.0uF capacitor to ground.
Pins: V29, T29, P29 and L25, L26, L27

It is recommended to follow the EVM decoupling strategy as it has been validated. Deviations are not recommended without performing static/dynamic PDN analysis to ensure targets (Reff, Cap LL, Impedance) are met. In some situations, the EVM may use 3-terminal capacitors for decoupling (due to low inductance packaging and performance.) Note how the 3-terminal capacitors are connected in schematic (for decoupling) - do not incorrectly implement them as an in-line/filter component.

Refer to the device-specific data sheet or PDN application note for the recommended power delivery targets. Use low ESL capacitors and 3-terminal capacitors connected with short traces to keep the mounting inductance very low.

Each I/O has an associated supply voltage used to power its I/O cell (VDDSHVx). This can be found in the pin list table of the device manual. If a VDDSHVn power rail is sourced from 3.3V supply, all IO on that rail must operate at 3.3V levels. If a VDDSHVn power rail is sourced from a 1.8V supply, all IO on that rail must operate at 1.8V levels.

Some interfaces span multiple VDDSHVn domains (GPMC for example). If using one of these interfaces, ensure all the VDDSHVn domains supporting that interface are source at the same voltage level.

Check that the supply voltage of all peripherals connecting with processor's I/Os, and verify the signal voltage level matches the corresponding I/O power domains (VDDSHVn). Connecting a signal to the wrong voltage level can cause leakage between the domains of the device.

Check that the supply voltage of all pull-up resistors that are connected to processor's I/Os matches the voltage applied to the corresponding I/O power domains (VDDSHVn). Pulling a signal to the wrong I/O voltage can cause leakage between the I/O domains of the device.

Most processor I/Os are not fail-safe. This means that voltage must never be applied to the I/Os while the corresponding I/O power domain (VDDSHVn) is off. Fail Safe IO Include: WKUP_I2C0_SCL, WKUP_I2C0_SDA, MCU_I2C0_SCL, MCU_I2C0_SDA, EXTINTN, USB0_VBUS (if supporting external circuit)

All other IO are NOT fail safe.

Refer to the device-specific data sheet for the recommended power sequencing requirements. If using an pre-approved PDN for the device (checklist item), then power sequencing should be compliant the device recommendations.

All power pins are required to be supplied proper voltage. Exceptions include VPP_MCU and VPP_CORE, which can be left open if features are unused. If using ADC logic, make sure REFP and REFN are supplied correct reference voltage. If unused, these pins should be left open.

VPP should only be enabled after SoC is powered (MCU and/or MAIN), and disabled prior to system power-down. It should never be powered without corresponding CORE logic supply. Alternately, VPP power can be enabled only when eFuses programming is performed.

In general, the following MCU voltage rails & domains must be supported on independent power rails supplied from a power resource that continues to operate when Main voltage domains are powered-down.

MCU Only Voltage Rails (Domains):

1. VDD_MCU_0V85 (vdd_mcu, vddar_mcu, vdd_mcu_wake1)
2. VDA_MCU_1V8 (vdda_mcu_pllgrp0, vdda_mcu_temp, vdda_mcu_adc0/1, vdda_por_wkup, vdda_wkup)
3. VDD_MCUIO_1V8 or VDD_MCUIO_3V3 (vddshv0_mcu, vddshv1_mcu, vddshv2_mcu)

In general, the following voltage rails & domains must be supported on independent power rails supplied from a power resource that continues to operate when MCU or MAIN domains are powered-down.

IO Retention mode for MCU Domain:

VDD_MCU_WAKE1 (L28)

VDDSHV0_MCU (H29, J28, K29)

IO Retention mode for MAIN Domain:

VDD_WAKE0 (U29)

VDDSHV2 (T30, T32, U31)

PMIC_WAKE pins are used to alter PMIC of status change and wake the device. The WAKE signals are active low polarity, and open drain (Requires pull-up to PMIC IO voltage). (Note pin polarity may not be labeled correctly).

MCU Wake: PMIC_WAKE1 (M33)

Main Wake: PMIC_WAKE0 (AJ34)

In general, the following Main & external SDRAM voltage rails & domains must be supported on independent power rails supplied from a power resource that continues to operate when all other Main & MCU voltage domains are powered-down.

S2R Voltage Rails (Device: Domains) for LPDDR4 mode of operation:

1. VDD1_LP4_1V8 (SDRAM: vdd1)
2. VDD_DDR_1V1 (SoC: vdd_ddr, vdd_ddr_c; SDRAM: vdd2, vddq)

The following IO requirements must be used to ensure DDR operation before/after entering low power mode:

1. DDR_RET_1V1 control input (pin AC8, G8, L8, G27) must be asserted (High) prior entering low power mode
2. DDR_RET_1V1 control input must be de-asserted (Low) after returning from low power mode

Refer to the device-specific power solution application note for the latest power management IC part numbers.

Verify WKUP_I2C0 is correctly connected to PMICs.

For TI PMIC solutions, signals are often powered by the VIO rails and IO type is Open-drain. Verify I2C signals are pulled to correct voltage domain.

Verify INT is correctly connect to between PMIC and processor IO (typically MCU GPIO)

For TI PMIC solutions, signal is often powered by the VIO rail and IO type is Open-drain. Verify IO is pulled to correct VDDSHVn for which IO is assigned.

Review the specific PDN to ensure the power rails are correctly assigned. Confirm the voltage levels are also correct for each PMIC.																																								
Review the specific PDN to ensure the GPIO are correctly assigned. Confirm the IO voltage levels are also correct for each PMIC.																																								
This means that the feedback inputs of the PMIC (the SMPSx_FDBK positive ones, as well as the SMPSx_FDBK_GND inputs where available) are connected through separate conductors to the points of load. This cannot always be obvious on the schematic (it's actually a matter of PCB routing), therefore confirm with customer that they are aware of the concept.																																								
A voltage divider network may be required to condition input voltage level to within PMIC remote monitoring specs, see PMIC datasheet for details.																																								
Refer to TI.com for additional information on Power Management IC design check items.																																								
Signal is usually controlled from the PMIC Reset Output (NRSTOUT) or other on-board logic. Pin for MCU_PORz: K32																																								
Signal voltage level is sourced from VDDA_WKUP, which is 1.8V. Often times reset signals are open-drain, make sure pulled to VDDA_WKUP.																																								
Signal is usually controlled from the PMIC GPIO (SOC_PORz) or combined with MCU_PORz if separate reset control is not required. Pin for PORz: P33																																								
Signal voltage level is sourced from VDDA_WKUP, which is 1.8V. Often times reset signals are open-drain, make sure pulled to VDDA_WKUP.																																								
If an non-TI PMIC reset controller is used, it must meet the requirements for pulse length and must be held low during power sequencing. It should also be held low while source clocks are starting. See data manual for specific timings.																																								
Warm reset for MCU and Main domains: Pin for MCU_RESETz: G36 Pin for RESET_REQz: F34																																								
Signal voltage level is sourced from VDDSHV0_MCU, which can be either 3.3V or 1.8V. Often times reset signals are open-drain, make sure pulled to same level as VDDSHV0_MCU.																																								
MCU Reset Output Pin (MCU_RESETSTATz): F36 Reset Output Pin (RESETSTATz): AL38																																								
MCU_RESETSTATz signal voltage level is sourced from VDDSHV0_MCU, which can be either 3.3V or 1.8V. RESETSTATz signal voltage level is sourced from VDDSHV0, which can be either 3.3V or 1.8V.																																								
Verify that Reset output(s) (RESETSTATz) has a pull-down of a few kOhm to ground.																																								
A clock source for WKUP_OSC0 is required for proper operation. T38 (WKUP_OSC0_XI), U37 (WKUP_OSC0_XO)																																								
See latest TRM/DM for supported clock frequencies and BOOT pin settings. ('W' represents MCU_BOOTMODE pin strapping)																																								
<table><tr><td>W2</td><td>W1</td><td>W0</td><td>Ref Clock</td><td>W2</td><td>W1</td><td>W0</td><td>Ref Clock</td></tr><tr><td>0</td><td>0</td><td>0</td><td>19.2MHz</td><td>1</td><td>0</td><td>0</td><td>26MHz</td></tr><tr><td>0</td><td>0</td><td>1</td><td>20MHz</td><td>1</td><td>0</td><td>1</td><td>27MHz</td></tr><tr><td>0</td><td>1</td><td>0</td><td>24MHz</td><td>1</td><td>1</td><td>0</td><td>Rsv</td></tr><tr><td>0</td><td>1</td><td>1</td><td>25MHz</td><td>1</td><td>1</td><td>1</td><td>No PLL Setup</td></tr></table>	W2	W1	W0	Ref Clock	W2	W1	W0	Ref Clock	0	0	0	19.2MHz	1	0	0	26MHz	0	0	1	20MHz	1	0	1	27MHz	0	1	0	24MHz	1	1	0	Rsv	0	1	1	25MHz	1	1	1	No PLL Setup
W2	W1	W0	Ref Clock	W2	W1	W0	Ref Clock																																	
0	0	0	19.2MHz	1	0	0	26MHz																																	
0	0	1	20MHz	1	0	1	27MHz																																	
0	1	0	24MHz	1	1	0	Rsv																																	
0	1	1	25MHz	1	1	1	No PLL Setup																																	
OSC1 Pins: P38 (OSC1_XI), N37 (OSC1_XO)																																								
If unused, OSC1_XI pins should be pulled to GND and OSC1_XO pins should be left unconnected/open.																																								

WKUP_LF_CLKIN Pin: M34

WKUP_LF_CLKIN is a standard LVCMOS IO and is shared with other pin functions. The input signal voltage level must match the supplied power to the IO domain (VDDSH0_MCU)

See device specific data manual for crystal requirements.

See device specific data manual for crystal requirements.

See device specific data manual for crystal requirements.

BOOTMODE and MCU_BOOTMODE pins do not have internal pull-up or pull-down resistors that are active during power reset. For early designs, recommend that all bootmode pins be brought out to an optional PU/PD pair with pop/no-pop options depending on the default functionality.

See device specific technical reference manual for complete bootmode definition(s).

MCU_BOOTMODE[2:0] Pins: J38, H36, G38

See device specific technical reference manual for complete bootmode definition(s).

MCU_BOOTMODE[5:3] Pins: J35, J34, H38

MCU Only (0 = normal, 1 = MCU only). See device specific technical reference manual for complete bootmode definition(s).

MCU_BOOTMODE[6] pin: H37

See device specific technical reference manual for complete bootmode definition(s).

MCU_BOOTMODE[7] Pins: K37

See device specific technical reference manual for complete bootmode definition(s). For GP, we recommend they set to 11 (boot) and for Safety enabled systems we recommend 01 (PBIST in parallel).

MCU_BOOTMODE[9:8] Pins: K38, J37

See device specific technical reference manual for complete bootmode definition(s).

BOOTMODE[0] Pins: B33

See device specific technical reference manual for complete bootmode definition(s).

BOOTMODE[3:1] Pins: D34, D33, B32

See device specific technical reference manual for complete bootmode definition(s).

BOOTMODE[6:4] Pins: N34, M36, M37

See device specific technical reference manual for complete bootmode definition(s).

BOOTMODE[7] Pins: M34

Boot values are latched at the release of power-on reset. If the bootmode pins are redefined for another purpose during operation, they must be released/set back to the proper levels to select the boot mode whenever the device enters the power-on reset state. This is specifically a concern if signal is driven from external peripheral.

If High Secure device, recommend UART boot as backup boot mode so keywriter can be loaded externally. Also requires MCU boot pins be accessible during manufacturing / key programming flow. See DM and TRM for additional specifics on Booting and UART boot.

P38 (WKUP_P30_SQ0_MCU), W37 (OSQ1[0:7])

Y38, Y34, V34... (MCU_ADC1_AIN[0:7])

ADC Reference pins (R35, AA35 and U35, W35) should be connected to same supply as VDDA_ADCx and GND, respectively (not open)

AC8 (DDR0_RET), G8 (DDR1_RET), L8 (DDR2_RET), G27 (DDR3_RET)

G37 (TRSTn)

G36 (MCU_RESETz)

G35 (TCK), AL36 (TMS), AL37 (TDI), H34 (EMU1), F35 (EMU0)

N33 (WKUP_I2C0_SCL), N35 (WKUP_I2C0_SDA), M35 (MCU_I2C0_SDA), G34 (MCU_I2C0_SCL), AN36 (I2C0_SDA), AP37 (I2C0_SCL)

AN35 (EXTINTN)

AA31 (VPP_CORE), L29 (VPP_MCU), AJ7 (MMC0_CALPAD)

A38, N13, N14, P13, P14, U15, U16, U17, U18, V15, V16, V17, V18, W15, W16, W17, W18, Y15, Y16, Y17, Y18, AA15, AA16, AA17, AB15, AB16, AB17, AB18, AV38

H33, J33, J31, AM14, AN13, G30, H32, K30, L30 (RSVD), AK2 (MMC0_VCTRL_TP)

AL23, AM26 (DSIO_ATB), AL24, AL25 (DSI1_ATB), AL27, AM29 (CSIO_ATB), AK29, AK31 (CSI1_ATB), AL30, AM33 (CSI2_ATB), AN (DPO_AUX_ATB), AM12, AM10, AM9, AL11, AM22, AM21, AM16, AM17 (SERDESx_ATB), Y7, AF7, H12, H14, N7, T7, G22, G17 (DDRx_ATB)
Assuming standard dual bank memory device is used, clock pins will connect to both Bank 0 and Bank 1 for each LPDDR4 interface. See Jacinto7 LPDDR4 Design Guide for complete description of connections for all LPDDR4 configurations. Each LPDDR4 interface is independent (no shared signals). DDR0 pins: AC1 (DDR0_CKP), AB2 (DDR0_CKN) DDR1 pins: B10 (DDR1_CKP), A11 (DDR1_CKN) - (If DDR1 not supported, pins can be left open) DDR2 pins: L1 (DDR2_CKP), K1 (DDR2_CKN) - (If DDR2 not supported, pins can be left open) DDR3 pins: A24 (DDR3_CKP), B25 (DDR3_CKN) - (If DDR3 not supported, pins can be left open)
Assuming standard dual bank memory device is used, clock pins will connect to both Bank 0 and Bank 1 for each LPDDR4 interface. See Jacinto7 LPDDR4 Design Guide for complete description of connections for all LPDDR4 configurations. Each LPDDR4 interface is independent (no shared signals). DDR0 pins: AD2, AC5, AB4, AC4, AB3, AC3 (DDR0_CA[0:5]) DDR1 pins: F12, C12, B12, C11, D12, E10 (DDR1_CA[0:5]) - (If DDR1 not supported, pins can be left open) DDR2 pins: K3, L3, K5, L4, K4, L7 (DDR2_CA[0:5]) - (If DDR2 not supported, pins can be left open) DDR3 pins: D25, B23, D24, C24, E23, F23 (DDR3_CA[0:5]) - (If DDR3 not supported, pins can be left open)
Assuming standard dual bank memory device is used, each CKE pins will connect to both Bank 0 and Bank 1. CKE0 is used for Bank 0 solution, CKE1 is used only if 2nd die is stacked. See Jacinto7 LPDDR4 Design Guide for complete description of connections for all LPDDR4 configurations. Each LPDDR4 interface is independent (no shared signals). DDR0 Pins: AB6 (DDR0_CKE0), AD3 (DDR0_CKE1, Can be left unconnected if not used) DDR1 Pins: D11 (DDR1_CKE0), C10 (DDR1_CKE1) - (If DDR1 not supported, pins can be left open) DDR2 Pins: L6 (DDR2_CKE0), J2 (DDR2_CKE1) - (If DDR2 not supported, pins can be left open) DDR3 Pins: C25 (DDR3_CKE0), G24 (DDR3_CKE1) - (If DDR3 not supported, pins can be left open)
Assuming standard dual bank memory device is used, CSN0 pins will connect to both Bank 0 and Bank 1 (separate connection for each bank). See Jacinto7 LPDDR4 Design Guide for complete description of connections for all LPDDR4 configurations. Each LPDDR4 interface is independent (no shared signals). DDR0 pins: AD7 (DDR0_CSN0_0 for Bank 0), AC7 (DDR0_CSN0_1 for Bank 1) DDR1 pins: E11 (DDR1_CSN0_0), G11 (DDR1_CSN0_1) - (If DDR1 not supported, pins can be left open) DDR2 pins: J3 (DDR2_CSN0_0), J6 (DDR2_CSN0_1) - (If DDR2 not supported, pins can be left open) DDR3 pins: G23 (DDR3_CSN0_0), G25 (DDR3_CSN0_1) - (If DDR3 not supported, pins can be left open)
Assuming standard dual rank memory device is used, CSN1 pins will connect to both Bank 0 and Bank 1 (separate connection for each bank). See Jacinto7 LPDDR4 Design Guide for complete description of connections for all LPDDR4 configurations. Each LPDDR4 interface is independent (no shared signals). DDR0 pins: AB7 (DDR0_CSN1_0 for 2nd rank of Bank 0), AD6 (DDR0_CSN1_1 for 2nd rank of Bank 1) DDR1 pins: F10 (DDR1_CSN1_0), G12 (DDR1_CSN1_1) - (If DDR1 not supported, pins can be left open) DDR2 pins: J7 (DDR2_CSN1_0), K7 (DDR2_CSN1_1) - (If DDR2 not supported, pins can be left open) DDR3 pins: F25 (DDR3_CSN1_0), E24 (DDR3_CSN1_1) - (If DDR3 not supported, pins can be left open)
Data interface does NOT support any bit or byte swapping. See Jacinto7 LPDDR4 Design Guide for complete description of connections for all LPDDR4 configurations. DDR0 pins: U1, V1 (DDR0_DQS0P,N), V3 (DM0), V6, U5, V7... (DQ[7:0]) DDR1 pins: A16, A17 (DDR1_DQS0P,N), E17 (DM0), D17, B17, C16... (DQ[7:0]) DDR2 pins: T1, R1 (DDR2_DQS0P,N), T2 (DM0), R7, T5, P5... (DQ[7:0]) DDR3 pins: A18, A19 (DDR3_DQS0P,N), E18 (DM0), F19, G19, E20... (DQ[7:0]) Note: If DDRx interface not supported, many pins can be left open. See unused signals for more details.
Data interface does NOT support any bit or byte swapping. See Jacinto7 LPDDR4 Design Guide for complete description of connections for all LPDDR4 configurations. DDR0 pins: AA1, Y1 (DDR0_DQS1P,N), AA4 (DM1), AA7, Y5, Y4... (DQ[15:8]) DDR1 pins: A13, A14 (DDR1_DQS1P,N), C15 (DM1), D14, E14, G13... (DQ[15:8]) DDR2 pins: N1, M1 (DDR2_DQS1P,N), M6 (DM1), M3, M4, N5... (DQ[15:8]) DDR3 pins: A21, A22 (DDR3_DQS1P,N), D21 (DM1), C20, B20, B21... (DQ[15:8]) Note: If DDRx interface not supported, many pins can be left open. See unused signals for more details.
Data interface does NOT support any bit or byte swapping. See Jacinto7 LPDDR4 Design Guide for complete description of connections for all LPDDR4 configurations. DDR0 pins: AF1, AE1 (DDR0_DQS2P,N), AG2 (DM2), AE4, AE3, AF4... (DQ[23:16]) DDR1 pins: A8, A9 (DDR1_DQS2P,N), D8 (DM2), C7, B7, B8... (DQ[23:16]) DDR2 pins: H1, G1 (DDR2_DQS2P,N), G4 (DM2), F2, G5, H3... (DQ[23:16]) DDR3 pins: A26, A27 (DDR3_DQS2P,N), C28 (DM2), D27, E27, G26... (DQ[23:16]) Note: If DDRx interface not supported, many pins can be left open. See unused signals for more details.

Data interface does NOT support any bit or byte swapping. See Jacinto7 LPDDR4 Design Guide for complete description of configurations for all LPDDR4 configurations.
DDR0 pins: AJ1, AH1 (DDR0_DQS3P,N), AJ5 (DM3), AG6, AH7, AH6... (DQ[31:24]) DDR1 pins: A3, A4 (DDR1_DQS3P,N), C1 (DM3), C3, C6, C5... (DQ[31:24]) DDR2 pins: E1, D1 (DDR2_DQS3P,N), D5 (DM3), D4, E6, E7... (DQ[31:24]) DDR3 pins: A29, A30 (DDR3_DQS3P,N), E30 (DM3), D30, B30, C29... (DQ[31:24]) Note: If DDRx interface not supported, many pins can be left open. See unused signals for more details.
CAL pin must be connected to GND through 240-ohm resistor. Each DDR interface must have its own/separate resistor. Each interface is independent (no shared signals). DDR0: AE8 (DDR0_CAL0) DDR1: G14 (DDR1_CAL0) DDR2: U7 (DDR2_CAL0) DDR3: F18 (DDR3_CAL0)
If supporting dual rank (stacked-die), CSN1_0, CS1_1, and CKE1 will be used to connect to 2nd rank. If single rank, these pins can be left open.
DDR_RET signal MUST be low ('0') for normal operation. If unused, connect to GND through resistor (range 0 to 1K-ohm). If supported, signal will connect to PMIC/External control. When signal is high ('1') the LPDDR4 interfaces are isolated from the processor and will only operate in self-refresh mode. DDR0 Pin: AC8 (DDR0_RET) DDR1 Pin: G8 (DDR1_RET) - (If DDR1 not supported, pin should be connected to GND thru resistor.) DDR2 Pin: L8 (DDR2_RET) - (If DDR2 not supported, pin should be connected to GND thru resistor.) DDR3 Pin: G27 (DDR3_RET) - (If DDR3 not supported, pin should be connected to GND thru resistor.)
Processor RESETN signal should connect directly to the memory's reset input. To help ensure signal is held low during power-up initialization, an optional pull-down resistor can be added. DDR0 Pin: AD5 (DDR0_RESETN) DDR1 Pin: G10 (DDR1_RESETN) DDR2 Pin: J5 (DDR2_RESETN) DDR3 Pin: C23 (DDR3_RESETN) TBD (DDR1_RESETN) - (If DDR1 not supported, pin can be left open)
The memory's on-die termination (ODT) should each be connected to VDD2 through 10K-ohm resistor. Pins: G2 (ODT_CA_A), T2 (ODT_CA_B)
The memory's IO calibration pin should each be connected to VDDQ through 240-ohm resistor. Pins: A5 (ZQ0), A8 (ZQ1)
Processor does NOT support for LPDDR4x (0.6v IO). Only standard LPDDR4 (1.1V IO) is currently supported.
It is very important to follow the DDR routing guidelines in the Jacinto7 LPDDR4 Design Guide (available on TI.com). These guidelines are very important to ensure a proper DDR operation. Failure to follow these guidelines will result in a nonfunctional DDR interface.
Always provide adequate decoupling capacitors on the DDR power rails both at the processor and at the memory device. Insufficient decoupling will lead to supply noise and will impact LPDDR4 functionality. Recommended following TI's EVM for decoupling strategy.
DDRSS0, DDRSS1, DDRSS2, DDRSS3 should always be used in incrementing order. E.g., if a single LPDDR component is used it should be connected to DDR0_* interface. If two LPDDR components are used they should be connected to DDR1_* and DDR2_* interfaces. Three interfaces should use DDR0_*, DDR1_*, DDR2_*.
The processor clock should connect to memory's clock input pin. It is recommended to have weak (100k-ohm) pull-down resistor optional to have series termination 0-ohm resistor (place near processor pin). This value can later be changed if signal integrity issues are identified. MCU_OSPIO_CLK: E32 MCU_OSPI1_CLK: F32 (If OSPI1 not supported, pin can be left open)
The processor chip select output pin(s) connect to memory's input chip select pin(s). It is recommended to have weak (10k-ohm) pull-up resistor on each chip select connection. This helps ensure the memory device is NOT selected unintentionally. OSPIO Pins: A32 (MCU_OSPIO_CSN0), A33 (MCU_OSPIO_CSN1), B34 (MCU_OSPIO_CSN2), C32 (MCU_OSPIO_CSN3) OSPI1 Pins: G32 (MCU_OSPI1_CSN0), G33 (MCU_OSPI1_CSN1) - (If OSPI1 not supported, pins can be left open) Note: Only CSN0 and CSN1 are supported for booting.

The processor data pins are bi-directional and connect to memory's data pin(s). The interface does not support bit swapping of data bits, these must be connected in the correct order. It is recommended to have weak (10k-ohm) pull-up resistor on D2, as memory devices also use this pin for write protection.

MCU_OSPI0_D Pins: B33 (D0), B32 (D1), C33 (D2), C35 (D3), D33 (D4), D34 (D5), E34 (D6), E33 (D7)

MCU_OSPI1_D Pins: E35 (D0), D31 (D1), G31 (D2), F33 (D3) - (If OSPI1 not supported, pins can be left open)

If data strobe IS supported, the processor's DQS input pin should be connected directly to the memory's DQS output pin. The processor's LBCLKO pin can be left unconnected/open. If data strobe is NOT supported, the processor's clock loop-back pin (LBCLKO) should be connected to processor's DQS input. The length of the connection should match the routing distance to the memory. See DM and TRM for additional PCB design specifics.

It is recommended to add a weak (100k-ohm) pull-down resistor on DQS. It is optional to have series termination 0-ohm resistor placed near the source (memory DQS or LBCLKO). This value can later be changed if signal integrity issues are identified.

Pin: C34 (MCU_OSPI0_DQS), D32 (MCU_OSPI0_LBCLKO)

Pin: F31 (MCU_OSPI1_DQS), C31 (MCU_OSPI1_LBCLKO)

If booting from serial flash memory, ensure the processor's selected chip select is supported for booting. CS0n is supported for as well as possible additional chip selects. See DM and TRM for additional boot details on chip selects.

When booting from serial flash, LBCLKO will be automatically driven as output - regardless if memory supports DQS. For this reason, make sure the LBCLKO pin is not used for auxiliary function, which could be impacted with LBCLKO output during boot period.

Regarding support for Serial flash memories, see article FAQ on E2E: <https://e2e.ti.com/support/processors/f/791/t/946418>

The following devices have been tested: See TI's EVM(s) for example devices

If unclear on support for specific OSPI memory, please post question on E2E

The processor OSPI memory rails is VDDSHV1_MCU, and support either 3.3V or 1.8V. This voltage is required to match the memory's voltage. Most designs use 1.8V serial flash interface, as it can often operate at higher data rate with lower voltage.

Some memory manufacturers support dual QuadSPI devices, where two QuadSPI devices are connected together to make it look like an 8bit OSPI device. This is not supported on the OSPI interface, as it requires extra commands to be sent from the controller.

The following pins should have 499-ohms (1%) resistor connected to GND.

USB0_RCALIB (AN18)

The following pins should have 3K-ohms (1%) resistor connected to GND.

AN11 (SERDES0_REXT), AL9 (SERDES1_REXT), AL20 (SERDES2_REXT), AM19 (SERDES4_REXT)

The following pins should have 499-ohms (1%) resistor connected to GND.

AM24, AL22 (DSIO1_TXRCALIB), AM28, AL28, AM31 (CSIO1/2_RXRCALIB)

The processor supports two internal reference clocks to generate the following possible frequencies. (Note an external reference clock is available to provide frequency if needed.)

SGMII (1G), QSGMII (5G): 100MHz, no SSC

XAUI/SGMII (2.5G), USXGMII (10G): 156.25MHz, no SSC

DisplayPort/eDP (): 19.2MHz, with SSC

USB (5G), PCIe (2.5G, 5G, 8G): 100MHz, with SSC

A single Serdes interface can support a maximum of two (2) different protocols simultaneously (assuming the clocking and pin assignments can be supported).

The processor supports single USB2.0 interfaces (USB0).

USB2.0 Signals: USB0_DP (AP17), USB0_DM (AP16), USB0_ID (AN17), USB0_VBUS (AN15), USB0_DRVVBUS (multiple locations)

VBUS pins cannot be connected directly to external/connector VBUS, as IO are not 5V compliant. Recommend using voltage divider and/or current limiter to ensure IO requirements are met. See DM or EVM for recommended circuit. VBUS pins can be considered safe if using recommended external circuit (due to limited input current)

AN15 (USB0_VBUS)

Refer to the Jacinto7 High-Speed Interface Layout Guidelines (available on TI.com) for detailed recommendations for proper USB SERDES signal connection and routing. Add appropriate constraints or routing requirements to your schematic.

USB3.x interfaces include both the Super Speed (SS) signals and USB2.0 connections for compatibility with older USB devices. The processor includes one USB3.x instances, USB0 USB3.x Signals: USB0_SSTXP, USB0_SSTXN, USB0_SSRXP, USB0_SSRXN (See DM for pinmux options)
The processor includes two sets of USB3.x signals (SS) for each USB instance. This allow direct support of Type C interfaces, w reversal is supported. Ensure the correct USB3.x instance and correct cable mux signals are used. Example: USB0_SSTX1P, USB0_SSTX1N, USB0_SSRX1P, USB0_SSRX1N is signal grouping '1'. USB0_SSTX2P, USB0_SSTX2N, USB0_SSRX2P, USB0_SSRX2N is signal groupint '2' (for cable reversal). Do not mix cable groupi
DC-blocking capacitors are required for USB3 transmit and receive pairs. The capacitors should be placed close to the USB3 tr If a USB3 connector is used in the design, the receive pair will be connected directly to the connector. The DC-blocking caps fo receive pair are present on the device transmitting at the far end of the USB3 connector.
Due to malfunctioning micro-B to Type-A adapters, recommend avoiding micro-AB connectors. For Super-Speed host mode, recommend Type-A ro Type-C only. For Super-Speed Dual Role - recommended Type-C only.
Refer to the Jacinto7 High-Speed Interface Layout Guidelines (available on TI.com) for detailed recommendations for proper P SERDES signal connections and routing. Add appropriate constraints or routing requirements to your schematic.
The processor includes four PCIe instances, PCIe0, PCIe1, PCIe2, and PCIe3. Instances PCIe0 and PCIe1 each instance can supp four (4) data lanes. Instance PCIe2 and PCIe3 can support up to two (2) data lanes. There is a restriction on the total number available - maximum is eight (8) total data lanes. Thus 4x 2L instances or 2x 4L instances. If using only 1 data lane, the selecte must be for PCIe signals TX0/RX0.
If the design uses external clock, it should be connect to a PCIe compliant 100MHz differential clock. If using internal SoC clock output, 50-ohm to GND resistors should be place near SoC for both CLKP/CLKn clock signals. AU8, AU9 (SERDES0_REFCLK_P/N - SERDES0/PCIe Input/Non-Compliant Output Clock Signals) AV4, AV3 (SERDES1_REFCLK_P/N - SERDES1/PCIe Input/Non-Compliant Output Clock Signals) AV22, AV21 (SERDES2_REFCLK_P/N - SERDES2/PCIe Input/Non-Compliant Output Clock Signals) AV15, AV16 (SERDES4_REFCLK_P/N - SERDES4/PCIe Input/Non-Compliant Output Clock Signals) AP5, AP4 (PCIe_REFCLK0_P/N_OUT - PCIe Compliant Output Clock Signals) AN9, AN8 (PCIe_REFCLK1_P/N_OUT - PCIe Compliant Output Clock Signals) AN6, AN5 (PCIe_REFCLK2_P/N_OUT - PCIe Compliant Output Clock Signals) AP8, AP7 (PCIe_REFCLK3_P/N_OUT - PCIe Compliant Output Clock Signals)
DC-blocking capacitors are required for PCIe transmit and receive pairs. The the capacitors should be placed close to the PCIe transmitter. If a PCIe connector is used in the design (off board), the receive pair will be connected directly to the connector (blocking caps). The DC-blocking caps for the receive pair will be present on the far-end PCIe device.
Refer to the Jacinto7 High-Speed Interface Layout Guidelines (available on TI.com) for detailed recommendations for proper S SERDES signal connections and routing. Add appropriate constraints or routing requirements to your schematic.
Processor supports upto eight (8) SGMII interfaces, but speeds higher than 2.5Gbps are only supported on two (2) specific SGM (SGMII1, SGMII2). Refer to DM/TRM for additional information on supports the higher rates.
Each port of the CPSWx ethernet switch can be configured for either RGMII or SGMII, and is available on specific pins. These p cannot be interchanged unless is explicitly defined in the device pinmux. For example, if RGMII1 is used then SGMII1 cannot a used at same time, unless the pinmux provides alternate options for RGMII1 or SGMII1.
As documented in Errata, some SGMII PHYs (including Marvel 88Q11111) looks at preamble data and makes packet keep/disc decisions based on preable data of 0x50. See errata item i2362 for more details.
Interface clocks and internal clocks are available on pins for visibility/debug. (See DM for pinmux options/locations). If these pins/signals are not muxed in your design for other purposes, have test point on them to allow the monitoring of internal clock support of hardware and software debugging.
TI strongly recommends that all board designs contain at least a minimal JTAG port connection to test points or header for ear prototype debugging. The minimum connections are TCK, TMS, TDI, TDO, and TRSTn. JTAG routes and component footprints (the pull-down resistor on TRSTn and the pull-up resistor(s) on TMS and TCK) can be deleted in the production version of the b desired.
Emulators that support adaptive clocking must be implemented correctly using the RTCK output. If the pin-out of your emulat connector includes the RTCK signal, such as the CTI20 or the MIPI60, the TCK signal should be connected to a buffer for the So buffer for the RTCK pin. An example can be found in appendix B of the Emulation and Trace Headers Technical Reference Man

The pins TMS, TDO, TDI are powered by the VDDSHV0 domain. The pins for TCK, TRSTz, EMU0, and EMU1 are powered by the VDDSHV0_MCU domain. For proper operation of most emulators, these signals must be operating at the same voltage level. If requirement for your emulator, ensure that VDDSHV0 and VDDSHV0_MCU are both at either 1.8 V or 3.3 V. Ensure Pull-up re to correct VDDSHV domain.

If trace operation is needed, the TRC_x signals must be connected to the emulation connector. All TRC_x signals are pinmuxed other signals. If the trace connections are needed, other muxed interfaces on these pins may not be used. Routes for TRC_x signals for trace must be short and skew matched. Trace signals are on a separate power domain and can be at a different voltage from other JTAG signals. For more recommendations on TRC/EMU routing, refer to the Emulation and Trace Headers Technical Reference Manual. A similar summary of this information is available at XDS Target Connection Guide.

TI strongly recommends that all board designs contain at least a minimal UART port connection to test points or a header foot support early prototype debugging (terminal). The minimum connections are TX and RX. This port can be deleted in the production version of the board, if desired. If TI software compatibility is needed, UART8 is used for console on TI EVMs.

If HS devices are supported with Keywriter utility, it is recommended to have both MCU_UART and WKUP_UART pinned out for R5 and Cortex-M3 KeyWriter logs.

I2C interfaces require a pull-up resistor on both the data and the clock lines (x_SDA and x_SCL). TI's EVM(s) typically use 2.2K-ohm resistors, but required value is dependent on loading. Ensure the pull-up resistors connect to the correct I/O voltage rail (VDD or VDDSHV). See data manual for additional information.

The WKUP_I2C0, MCU_I2C0, I2C0, and I2C1 controllers have dedicated I2C compliant open drain buffers and standard/fast speed modes (up to 400 kbps in 1.8 V/3.3 V mode). All other I2C controllers are multiplexed with standard LVCMOS I/O, connected to emulate open drain, and support standard/fast mode. These 'emulated' I2C ports are not fully compliant with the I2C specification. Particular falling edges could be too fast.

See latest processor documentation for support of high speed mode(s).

Include a series resistor (default to 0-ohms) on MMCx_CLK placed as close to SoC as possible. This signal is used/looped back on read transactions, and the resistor may be needed to eliminate possible signal reflections, which can cause false clock transactions. It is important to default to 0-ohms, as the processor's IO are trimmed to 50-ohms output impedance, likely matching the PCB trace impedance.

Pull-up resistors are needed on all data and command signals (not clock). These pull-ups should be approx. 47K-50K-ohms for all embedded devices connections, such as eMMC. The pull-up resistors should be connected to same supply as the MMC I/O: VDDSHV0_MMC0 for MMC0, VDDSHV5 for MMC1.

The UHS-I devices support both high voltage (3.3V) and low voltage (1.8V) I/O. The switchable voltage output must be connected to proper IO voltage rail (VDDSHV5). Be aware it is possible for power source and memory device to get out-of-sync with respect to voltage setting. For a design example, TI's EVM(s) typically support dual voltage SD cards.

During boot, the ROM code checks the status of the card detect pin (SDCD, pin AE25). It must be low to indicate card detected. Booting from SD card. Note booting from eMMC device (MMC0) does not require SDCCD support.

While the IO voltage for SD Card interface can be either 1.8V or 3.3V, the card's VDD signal should be connected to a fixed 3.3V.

The processor includes two (2) Functional Safety monitor/error signals - one for each processor domain. If not used, these pins should be left open.

MCU Domain: N36 (MCU_SAFETY_ERRORN)

Main Domain: AM34 (SOC_SAFETY_ERRORN)

The processor supports voltage comparators for monitoring external voltages. These are often part of Functional Safety requirements. Each of the monitors is designed to monitor a fixed voltage range, and may or may not be able to support external resistor divider support monitoring of other voltages. Usage of the monitors is optional. Pins can be left open/unconnected if not used.

K28 (VMON1_ER_VSYS) nominal input is 0.45V. Can use voltage divider to adjust external voltage.

N27 (VMON2_IR_VCPU) Must be used to monitor the VDD_CPU voltage

J30 (VMON3_IR_VEXT1P8) nominal input is 1.8V. Cannot use voltage divider unless divided voltage is buffered.

P28 (VMON4_IR_VEXT1P8) nominal input is 1.8V. Cannot use voltage divider unless divided voltage is buffered.

R29 (VMON5_IR_VEXT3P3) nominal input is 3.3V. Cannot use voltage divider unless divided voltage is buffered.

TI's System Configuration Tool is an intuitive and comprehensive collection of graphical utilities for configuring pins, peripherals, and subsystems for processors. The tool can be used to help check some hardware design elements as well as accelerate software development by providing software files customized to your design. Many of TI's EVM design packages include the SysConfig tool. It can be used as an example/starting point for custom designs.

onal information can be found in the corresponding documenation: SPRAD91

Status	Comment/Notes
N/A	Jag har inte gjort någon estimering.
Verified	Jag använder SK-AM69 referensschema med modifikation
Verified	Det blir 0.8V
Verified	Det blir 0.85V
Verified	Det blir 1.8V
Verified	Det blir 1.8V
Verified	Det blir 1.8V
Verified	Det blir 1.8V
Verified	Det är 3.3V
Verified	Det är 1.8V
Verified	Det är 3.3V
Verified	Det är 0.8V
Verified	Det är VDD_CPU_AVS
Verified	Det är 0.8V
Verified	Hos SysConf så ska det vara 1.8V

Verified	Hos SysConf så ska det vara 1.8V
Verified	Hos SysConf så ska det vara 1.8V
Verified	Ja
Verified	Dom alla kommer från VDD_CORE_0V8
Verified	Ja ! 0.85V
Verified	Ja ! 0.85V
Verified	Det är 0.8V
Verified	Det är 0.8V
Verified	Det är 0.8V
Verified	Det är 1.8V
Verified	Det är 1.8V
Verified	Det är 1.8V
Verified	Det är 1.8V
Verified	Det är 1.8V
Verified	Det är 1.8V
Verified	Det är 1.8V
Verified	Det är 3.3V
Verified	Det är 3.3V
Verified	Det är 3.3V.
Verified	Det är 1.8V

Verified	Kan både vara 1.8V eller 3.3V beroende på inställning hos AND-grinden.
Verified	VPP har spänning
Verified	VPP har spänning
Verified	Implementerat !
Verified	Jag använder samma som SK-AM69
Verified	Jag använder samma som SK-AM69
Verified	Filterna ser OK ut för Analog Power
Verified	Jag har lagt ut så mycket kondensatorer och ferritpärlor som möjligt. Jag följer SK-AM69
Verified	Jag använder SK-AM69 referensschema
Verified	Dom finns
Verified	Jag använder SK-AM69 referensschema
Verified	Jag använde 0201 och inga genomkondensatorer. Detta har med att jag fick inte plats med genomkondensatorer.
Verified	Det gör dom alla. Som olika grupper med olika spänningsnivåer. Ingen är blandad.
Verified	Ja. Dom har unika spänningsnivåer. Väldigt enkelt att se deras spänningsnivå grundat på deras pin-namn
N/A	Förstår inte....

Verified	Det är rätt spänningsnivåer på alla IO
Verified	Jag använder PDN som finns för SK-AM69. Se 6.10.2 Power Supply Sequencing från databladet av AM69
Verified	Jag har 0 ohm resistorer innan VPP_MCU och VPP_CORE.
N/A	Vet inte vad detta är...
N/A	Jag använder inte detta
N/A	Jag använder inte detta
N/A	Jag använder inte detta
Verified	Jag använder Tis PMIC
Verified	
Verified	

Verified	
Verified	
Verified	
Verified	
Verified	
Verified	Har kopplat PMIC utgången till SoC
Verified	Ja ! En VDA_PLL_1V8 med 10 kOhm pull-up
Verified	Fixat ! Har även kopplat ihop med MCU_PORz
Verified	Ja ! En VDA_PLL_1V8 med 10 kOhm pull-up
N/A	Jag använder TI PMIC
Verified	Jag sätter dom med en pull-up på 10 kOhm
Verified	3.3V blir det ! VSYS_3V3
Issue/Open	Jag inväntar vilka enheter som behövs nollställas.
Issue/Open	
Verified	OK. 10 kOhm pull-down
Verified	19.2 MHz har jag valt.
Verified	Jag har satt pull-down på alla.
N/A	
Verified	Har satt den till GND och OSC1_XO som ECR

N/A	
N/A	
Verified	Jag använder SK-AM69 referensschema
Verified	Jag använder SK-AM69 referensschema
N/A	Jag använder kristall
Verified	Jag har satt pull-down på alla.
Issue/Open	
Issue/Open	
Issue/Open	
Issue/Open	
Issue/Open	
Issue/Open	
Issue/Open	
Issue/Open	
Issue/Open	
Issue/Open	
Issue/Open	
N/A	Stödjer en HS
Verified	OSC1_XI har en pull-down på 10 kOhm
Verified	Pull-down på 10 kOhm
Verified	DDR0 endast fick en pulldown
Verified	10 kOhm pull-down
Verified	10 Kohm pull-up
Verified	10 Kohm pull-up
Verified	1.8 kOhm pull-up
Verified	10 Kohm pull-up
Verified	10 kOhm pull-down för AJ7 för jag ska använda MMC0
Verified	Intresseklubben noterar...
Verified	Klart !

Verified	Klart !
Verified	
Verified	Den differentiella klockan är dragen nu
Verified	
Verified	AD2 OK. AC5 OK. AB4 OK. AC4 OK. AB3 OK. AC3 OK.
Verified	
Verified	AB6 OK. AD3 OK.
Verified	
Verified	AD7 OK. AC7 OK.
Verified	
Verified	AB7 OK. AD6 OK.
Verified	
Verified	U1, V1 OK. V3 OK. V6, U5, V7.
Verified	
Verified	

Verified	
Verified	AE8 OK
Verified	
Verified	
Verified	
Verified	
Verified	
Verified	
Issue/Open	
Issue/Open	
Verified	
Not Checked	
Not Checked	

Not Checked	
Not Checked	
Not Checked	
Not Checked	
Not Checked	
Not Checked	
Verified	
Verified	
Verified	
Issue/Open	?
Issue/Open	?
Not Checked	
Not Checked	
Not Checked	

[illegible]

[illegible]